Grammar–KT: Operationalising Grammar Knowledge Components for Knowledge Tracing

Abdullah Akram
University of Cambridge
aa2527@cam.ac.uk

Abstract

Knowledge tracing (KT) assumes that every assessment opportunity is linked to one or more knowledge components (KCs), yet grammar curricula commonly express learning targets as prose descriptors rather than operational skills. This report presents Grammar–KT, an auditable framework for constructing and comparing grammar KC representations. A frozen pilot design selects 139 descriptors from a digest-declared English Grammar Profile snapshot and uses constrained, two-phase model-assisted normalisation to map supported claims into a closed six-dimensional representation of English verbal morphosyntax. Deterministic realisation then constructs a KC-independent item bank. A separate structural oracle generates one fixed synthetic event stream before any evaluated KC policy is selected, so alternative representations are compared on identical items and outcomes. Development-only candidate diagnostics, frozen KC policies, Q-matrix construction, and non-updating compositional probes make support, coverage, and transfer assumptions explicit. The present revision documents the implemented methodology and its evidence requirements; quantitative research comparisons are deliberately reserved for a fresh, validated run of the final pipeline. The contribution is therefore a controlled experimental design for studying grammar representations, not a claim that synthetic performance establishes the cognitive reality of any KC.

1 Introduction

Knowledge tracing (KT) estimates a learner's changing knowledge from a chronological sequence of educational interactions (Corbett and Anderson, 1995; Abdelrahman et al., 2023). Although KT models vary in their assumptions and architectures, their inputs normally associate each item with one or more skills or *knowledge components* (KCs). A binary Q-matrix makes this content model explicit: its (i, k) entry records whether item i exercises KC k. This matrix is often treated as supplied data, but it already embodies a consequential hypothesis about the boundaries of knowledge.

That hypothesis is especially difficult to specify for open-ended language tutoring. The Teacher–Student Chatroom Corpus version 2 (TSCCv2), for example, contains 260 written English lessons, 41.4K conversational turns, and approximately 363K word tokens from two teachers and thirteen learners. Its annotations include discourse sequences, response relations, pedagogical focus, and grammatical-error corrections (Caines et al., 2022). These annotations make TSCCv2 a motivating future application for dialogue KT, but they do not turn every learner utterance into a predefined assessment item. A turn may contain explanation, scaffolding, elicitation, correction, or several grammatical phenomena at once. Neither the assessment opportunity nor its KC decomposition is therefore given by the corpus.

Dialogue-based KT makes this missing representation visible. Scarlatos et al. (2025) use large language models (LLMs) to identify KCs and correctness at individual turns in mathematics tutoring dialogues before applying KT models, with expert annotations used to assess the automatic labels. More recent work additionally models the difficulty of tutor-posed tasks (Huang et al., 2026). These studies operate in domains where an existing mathematical standard provides a comparatively explicit candidate skill space. Grammar raises an earlier question: how should that space itself be constructed?

Consider the erroneous form *Why didn't he went?*. A coarse analysis might assign one *past-question* KC. A factorised analysis might instead identify finite-past selection, DO-support, negation placement, and non-subject-WH formation. An interaction-sensitive analysis might additionally represent the dependency between DO-support and inversion. These alternatives share the same observed sentence but imply different learner states, opportunity counts, and Q-matrices. No decomposition is neutral.

Proficiency resources offer curriculum evidence but do not solve this problem directly. The Common European Framework of Reference for Languages (CEFR) organises proficiency through broad functional scales (Council of Europe, 2020), whereas the English Grammar Profile (EGP) associates more detailed grammatical competence statements with CEFR levels (O'Keeffe and Mark, 2017). An EGP descriptor may nevertheless leave a grammatical property unspecified, cover several alternatives, or instantiate the same structural configuration as a different descriptor. Treating each descriptor as a KC would conflate source wording, curriculum evidence, grammatical structure, and a cognitive modelling hypothesis.

Grammar–KT separates these layers. It first maps source descriptors into a closed canonical representation, then constructs concrete grammatical items without consulting a KC ontology. Synthetic learner outcomes are generated from a separate, declared structural oracle. Only after both the item bank and event stream have been fixed are candidate KCs selected or supplied, projected onto items, and evaluated. This order addresses an important confound: if a KC policy determines which items exist or how responses are generated, predictive differences between policies cannot be attributed to representation alone.

This report asks three questions:

RQ1 How reliably can EGP descriptors be normalised into a declared canonical representation of English verbal morphosyntax?

RQ2 Which candidate grammar distinctions are supported, identifiable, and retained using development-only structural evidence?

RQ3 How do frozen KC representations differ in Q-matrix coverage and transfer to unseen grammatical combinations under identical synthetic learner–item outcomes?

The main contributions are: (i) a conservative six-dimensional GRAMMARCELL representation with explicit uncertainty and source provenance; (ii) a fixed, split-independent controlled item bank; (iii) a development-only KC selection procedure that diagnoses nuisance sensitivity and observational equivalence; and (iv) an ontology-independent simulation and frozen-probe protocol for controlled representation comparison. These are methodological and software contributions. The study does not claim that the selected dimensions are cognitively atomic, that the synthetic learners approximate human acquisition, or that results transfer directly to conversational language learning.

The remainder of the report reviews the relevant literature, formalises the task and pilot data, describes the implemented pipeline and evaluation design, states the present evidence boundary, and discusses the interpretation and limitations of the framework.

2 Related Work

2.1 Proficiency resources and grammatical representation

The CEFR provides proficiency scales across language activities rather than a machine-readable inventory of grammatical skills (Council of Europe, 2020). The English Profile programme complements these scales with empirical descriptions of grammatical and lexical competence. The EGP draws on learner-corpus frequency, accuracy, and dispersion to associate grammar competence statements with CEFR levels (O'Keeffe and Mark, 2017). Its level assignments have also been revisited quantitatively using NLP methods (Verratti-Souto et al., 2025). The present work does not reassess CEFR levels. It instead asks how the grammatical content of a bounded EGP subset can be represented without converting prose descriptors directly into KCs or silently completing information that the source does not provide.

This distinction separates a *curriculum source* from a *domain model*. Multiple descriptors can support one grammatical configuration, one descriptor can explicitly support alternatives, and many descriptors do not fully determine all properties required by a closed ontology. Grammar–KT retains those cases as partial or out of scope rather than forcing them into an exact cell.

2.2 Knowledge components, Q-matrices, and model selection

Classical Bayesian Knowledge Tracing (BKT) models mastery as a latent binary skill state updated after each opportunity (Corbett and Anderson, 1995); Deep Knowledge Tracing replaces explicit Bayesian state transitions with a neural sequence model (Piech et al., 2015). Broader surveys show that modern KT architectures differ substantially in their state representations and use of item information (Abdelrahman et al., 2023). All nevertheless require a definition of the content whose learning is being traced.

KCs are not necessarily natural atoms waiting to be annotated. The Knowledge–Learning–Instruction framework treats them as explanatory units linking tasks, learning processes, and instruction (Koedinger et al., 2012). Learning Factors Analysis evaluates and refines a tutor's cognitive model using student data and explicit complexity trade-offs (Cen et al., 2006). Related factor models likewise express performance through skill difficulty and practice terms (Pavlik et al., 2009). These approaches motivate treating a KC inventory as a hypothesis rather than administrative metadata.

In cognitive diagnosis, the Q-matrix specifies which latent attributes are required by each item. Expert construction is subjective, and misspecification can affect learner classification; empirical validation and refinement methods therefore examine the adequacy of proposed entries (de la Torre and Chiu, 2016; Pardos and Dadu, 2018). Grammar–KT intervenes before human response data are available. It tests the structural support, scope, and identifiability of declared candidate rules, freezes a policy, and leaves later learner-data validation as a separate stage.

2.3 Knowledge tracing in tutoring dialogue

Open-ended tutoring dialogue does not arrive with question identifiers, correctness labels, or a fixed KC set. Scarlatos et al. (2025) frame tutor–student dialogue as a sequence of formative assessments and use LLMs to annotate turn-level KCs and correctness for CoMTA and MathDial. Their pipeline retrieves mathematics standards as candidate KCs and evaluates the automatic annotations against expert labels. Huang et al. (2026) extend conversational KT by explicitly modelling student ability and task difficulty through an LLM and item-response-theoretic representation.

Grammar–KT addresses a complementary, upstream problem. Rather than assigning a turn to an already available skill standard, it constructs and compares the skill representation itself. TSCCv2 provides a plausible eventual test bed because it includes learner language and grammatical corrections (Caines et al., 2022); however, applying the present ontology to TSCCv2 would still require a separate definition of assessment opportunities, context-sensitive correctness, and multi-phenomenon turn annotation.

2.4 Exercise generation for language learning

Prior work has generated educational questions conditioned on learner history or target difficulty (Srivastava and Goodman, 2021; Cui and Sachan, 2023). Grammar-oriented systems generate reading, multiple-choice, or gap-filling exercises from passages and examples (Chan et al., 2022; Bitew et al., 2023). Such work demonstrates both the cost of manual authoring and the importance of correctness, naturalness, and answer uniqueness.

The present item generator deliberately makes a different trade-off. It uses controlled transformations with fixed lexical frames so that the relationship among a canonical cell, a concrete surface derivation, and a later KC projection remains inspectable. This reduces lexical and discourse diversity but avoids using unconstrained generation to hide changes in the experimental content model.

2.5 Synthetic learners and controlled evaluation

Synthetic data can test whether a KT implementation recovers known patterns or responds to controlled changes (Pagonis et al., 2024). It is also easy to overinterpret. Recent evaluation of LLM-simulated students finds substantial behavioural and cognitive limitations even when linguistic output appears plausible (Scarlatos et al., 2026). Grammar–KT therefore uses a transparent numerical structural oracle rather than an LLM role-play student, and calls its outputs an experimental world rather than human learner data. The oracle is useful for testing leakage, component reuse, cold-start behaviour, and compositional transfer; it cannot establish the cognitive validity or acquisition order of grammar KCs.

3 Task Formulation and Pilot Data

3.1 From descriptors to candidate annotations

Let $D = \{d_1, \ldots, d_n\}$ be a set of source grammar descriptors. A conventional KT event is a tuple

$$e_t = (l_t, i_t, y_t, K(i_t)), \qquad (1)$$

where l_t is a learner, i_t is an item or assessment event, $y_t \in \{0, 1\}$ is the observed outcome, and $K(i_t)$ is the associated KC set. A prose grammar resource directly supplies neither i_t nor $K(i_t)$.

Normalisation maps a descriptor to a status and zero or more constrained grammar cells. For complete mappings, define

$$C(d) = \{c_1, \ldots, c_m\}, \qquad C : D \to \mathcal{P}(\mathcal{C}), \tag{2}$$

where each $c \in \mathcal{C}$ is an exact assignment over the declared dimensions. A descriptor can support multiple independently asserted cells, and several descriptors can converge on the same cell. Partial, unresolved, out-of-scope, and schema-failure mappings contribute no exact downstream cell; their retained records remain part of the attrition and reliability evidence. The canonical inventory is the content-deduplicated union

$$\mathcal{C}_D = \bigcup_{d \in D} C(d), \tag{3}$$

with separate source–cell edges preserving provenance.

For each exact cell, a deterministic grammar defines a set of admissible realisations $\mathcal{R}(c)$. A realisation specifies lexical frame, subject features, any WH role or imperative subtype, the surface answer, and named operations such as DO-support or operator inversion. The item constructor selects from this space using a fixed measurement design:

$$I = B(\mathcal{C}_D, \{\mathcal{R}(c) : c \in \mathcal{C}_D\}). \tag{4}$$

Crucially, the bank function B receives neither a KC policy nor a fold label. Each accepted item i exposes an opportunity x_i containing its cell, realisation, and operation evidence.

A KC policy π is a frozen set of activation rules. For a candidate KC k, let $a_{\pi k}(x_i) \in \{0, 1\}$ denote rule activation. The policy-specific annotation and Q-matrix are

$$K_\pi(i) = \{k : a_{\pi k}(x_i) = 1\}, \tag{5}$$

$$Q^\pi_{ik} = \mathbb{I}[k \in K_\pi(i)]. \tag{6}$$

This definition permits two items from the same canonical cell to differ when a rule depends on a surface operation, while purely cell-scoped rules remain invariant across realisations.

The simulation uses a separate mapping $O(i)$ from items to structural oracle features. It generates learner identities, item order, difficulty, and outcomes before π is selected. Alternative policies therefore annotate the same event stream

$$E = \{(l_t, i_t, y_t, d_{i_t}, s_t)\}_{t=1}^{T} \tag{7}$$

rather than defining different synthetic worlds. The distinction between $O(i)$ and $K_\pi(i)$ is the central internal-validity boundary of the experiment.

3.2 The GrammarCell representation

The pilot ontology covers single-clause English verbal morphosyntax. An exact GrammarCell is the ordered tuple

$$c = (\text{tense}, \text{aspect}, \text{voice}, \text{polarity}, \text{clause}, \text{modal}). \tag{8}$$

Table 1 gives the closed domains. The representation excludes lexical aspect, argument structure as a canonical dimension, contractions, semi-modals, embedded questions, non-BE passives, and multi-clause relations. These exclusions are modelling choices, not claims about which phenomena matter to language learners.

Two cross-field constraints prevent impossible combinations in the declared space. Imperatives take `tense=NA` and `modal=none`; a central modal occupies the finite position and likewise requires `tense=NA`. Lists and null values are permitted while representing source uncertainty, but only scalar assignments become exact canonical cells.

3.3 Frozen pilot source design

The external input is a consult-only parsed EGP snapshot declared to contain 1,222 JSONL records and identified by SHA-256. The source stage refuses a file with a different digest or record count. The frozen pilot contains 139 unique descriptors: 23 controls inherited from an earlier narrow regression suite and 116 fresh records selected in declared structural strata. Table 2 groups those strata. The design intentionally includes seven negative controls for constructions outside the canonical scope.

The 139 primary descriptors are expanded to 147 isolated annotation units by repeating eight descriptors. These repeats support explicit agreement diagnostics for result category, exact cell set, each dimension, Phase-2 routing, and downstream canonical contribution. They are a small reliability probe rather than a basis for a general inter-annotator agreement claim.

This is a purposive pilot sample, not a probability sample of the EGP. It was constructed to exercise targeted tense, aspect, voice, polarity, clause, modal, and out-of-scope cases. The repository contains an executable sampling utility, but its MAIN-study quota file remains deliberately unset. Accordingly, the present report makes methodological and within-pilot claims only.

Dimension	Closed values	Interpretation and principal exclusions
Tense	present, past, NA	Morphological finite tense. Future is not analysed as a tense; NA is required when an imperative or central modal occupies the relevant structural position.
Aspect	none, progressive, perfect, perfect_progressive	Grammatical aspect expressed by the auxiliary chain; lexical aspect and habituality are excluded.
Voice	active, passive	Canonical active/non-passive versus BE-passive voice; GET-passives and middle constructions are excluded.
Polarity	positive, negative	Clause-level polarity; lexical negation and negative concord are excluded.
Clause	declarative, polar_question, subject_wh_question, non_subject_wh_question, imperative	Supported matrix-clause types; question tags, embedded questions, and indirect questions are excluded.
Modal	none, can, could, may, might, must, shall, should, will, would	No central modal or one named central modal; semi-modals and periphrastic modal expressions are excluded.

Table 1: Closed domains of GRAMMARCELL version 1. A list or null can occur during normalisation, but it is uncertainty metadata rather than an exact canonical value.

Declared sampling stratum	Descriptors
Earlier regression controls	23
Present/past simple and aspect contrasts	50
Canonical BE-passives	14
Direct question forms	10
Imperatives	4
Clause-level negation	4
Nine central modals (three each)	27
Out-of-scope negative controls	7
Total unique descriptors	139
Repeated annotation units	8
Total isolated units	147

Table 2: Frozen pilot source design. Strata overlap conceptually, but the declared sampling metadata assigns each selected descriptor to one row.

4 Grammar-to-KT Method

Figure 1 shows the implemented order and the two separations that make the comparison interpretable. First, normalisation converts source claims into canonical content without consulting items or KCs. Second, item identities and synthetic outcomes are fixed before a KC policy is selected. The later policy intervention can consequently change annotations and learner state representations, but not the questions asked or the answers observed.

4.1 Constrained, two-phase normalisation

Every annotation unit is handled in an isolated model context and must return a single schema-valid record. Phase 1 exposes only the EGP identifier, supercategory, subcategory, guideword, and competence statement. Examples, CEFR level, other descriptors and mappings, running counts, item designs, and KT outputs are withheld. For each dimension, the rules assign an exact scalar when directly supported, separate cells for explicitly independent alternatives, a bounded list for a stated superclass, and null when the text provides no usable constraint. Lists never license their members, and the procedure does not manufacture Cartesian combinations.

Phase 2 is a bounded repair pass. Only a partial Phase 1 record is normally eligible, and its note must name the dimensions that examples may refine. Phase 2 receives the unchanged descriptor, Phase 1 record, and that descriptor's examples. A transition validator freezes exact and ineligible dimensions, preserves branch relationships, and requires every revised cell to descend from a Phase 1 cell. Examples can supply positive evidence for declared alternatives; frequency or absence cannot narrow the mapping. A direct contradiction turns the record into unresolved rather than silently broadening or replacing an exact value.

Both phases apply JSON-schema, domain, status, and cross-field checks. The artifact retains the complete prompt, raw response, parsed record, invocation metadata, attempt number, and validation errors. The primary and repeated units are executed independently. Reliability is reported separately for result status, exact cell set, individual dimensions, Phase-2 routing, and canonical contribution; a single headline coefficient would conceal which decision disagreed.

Complete mappings are serialised in the schema's fixed dimension order and assigned content-derived identifiers. Equal tuples therefore deduplicate regardless of descriptor identity or processing order. A separate edge table retains every

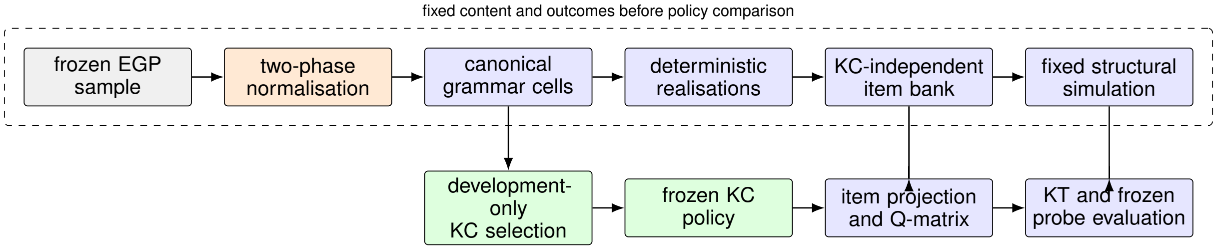

Figure 1: The implemented experimental order. Orange denotes isolated model-assisted judgements, green denotes the ontology intervention, and blue denotes deterministic or fixed-data stages. The simulator uses its own structural oracle, never an evaluated KC policy.

source-to-cell path, so deduplication does not discard many-to-one provenance.

4.2 Deterministic realisation and fixed items

The realiser combines an exact cell with one of ten declared predicate frames, subject person and number, and any WH role or imperative subtype. It first constructs the ordered verbal chain

$$\mathcal{V} = [\text{MODAL}][\text{PERFECT HAVE}] \\ [\text{PROGRESSIVE BE}][\text{PASSIVE BE}]\ \text{MAIN}, \tag{9}$$

then assigns finiteness and agreement and applies deterministic rules for negation, operator inversion, DO-support, WH placement, and imperatives. The retained realisation record includes its tokens, finite site, auxiliary chain, lexical conditions, and named operations. Invalid cell–frame combinations are rejected explicitly.

The fixed bank design creates a baseline opportunity for every realisable cell and adds declared coverage for source-licensed imperative subtypes, WH roles, lexical-versus-copular operator contrasts, and representative agreement conditions. Item identifiers are derived without a KC label or experimental fold. Each item is a controlled transformation with fixed lexical material and a deterministic target answer. Fold membership is joined only when an experimental stage needs it; it cannot influence whether an item exists.

Deterministic validation checks record shape, item identity, provenance, surface derivation, punctuation, uniqueness, and target agreement. A separate model diagnostic then records five binary judgements: structural plausibility, naturalness, dependence on world knowledge, unsupported construction, and suspected answer ambiguity. Acceptance requires the declared target pattern across all five fields. Five calls are repeated to expose diagnostic instability. This automated gate is a reproducible screening device, not a substitute for expert item review.

4.3 Development-only structural KC selection

The structural selector starts from a versioned hypothesis family containing canonical rules at several granularities, surface-operation rules, nine per-modal alternatives, and seven declared pairwise interactions. Marked obligations cover finite present and past, perfect and progressive structure, passive voice, negative polarity, supported non-declarative clauses, and each central modal. Active, positive, declarative, no-modal, and no-aspect values are treated as background under this policy. This markedness declaration is an explicit modelling prior rather than a linguistic universal.

All discovery and selection use development cells only. The selector expands each development cell over a deterministic nuisance-realisation grid. A rule that is invariant across the grid is diagnosed as cell-scoped; a rule that changes with lexical frame or surface operation is realisation-scoped. A mixed operation rule is retained as useful item evidence but is not allowed to stand in for a cell-level coverage obligation. Candidates must also meet minimum cell and descriptor support. An interaction needs at least two development cells, an activation set strictly narrower than both parents, and a declared residual structural fact.

Candidates with identical development activation columns are observationally equivalent for this sample. The implementation collapses each such class and chooses a deterministic representative; that tie-break is bookkeeping, not empirical evidence for one interpretation. Eligible candidates must jointly cover every development cell, every salient marked fact, and declared Hamming-one or minimal structural contrasts. A lexicographic greedy pass constructs a feasible cover, after which backward deletion removes every redundant selected rule. The result is therefore inclusion-minimal under the declared obligations, but is not claimed to be a globally smallest ontology. The selected rules and trace are written as a frozen ordinary policy before either holdout is evaluated.

4.4 KC materialisation and Q-matrix construction

Once policy π is frozen, its rules are applied to each concrete item opportunity. This preserves an important scope distinction: two items from one cell can differ under an operation rule, whereas a cell rule must activate identically. The Q-matrix is a mechanical conversion of this projection, not an additional inference stage.

Integrity checks reject duplicate identities, unknown cells or KCs, missing item projections, and disagreement between the edge list and dense matrix. Zero-KC rows are deliberately retained for inductive exact-cell controls on unseen cells. Density, row-width distribution, duplicate columns, low-support columns, uncovered rows, and cell-versus-realisation edge counts are reported as scientific diagnostics instead of being repaired silently.

5 Experimental Design

5.1 Canonical fold and ontology conditions

The canonical fold is declared independently of the item bank. It assigns all 24 cells in the reference inventory explicitly: 16 development cells, seven compositional holdouts, and one novel-feature holdout. Exact-inventory validation prevents a new or changed cell from defaulting silently into the development set. In a compositional holdout, every salient marked fact occurs in at least one development cell but their conjunction is new. The novel-feature holdout contains at least one salient fact absent from development. These definitions concern canonical content, not random item rows.

Table 3 lists the six representations to report. They answer different questions and are therefore labelled by their access to prior information. The selected structural, factorised-development-frozen, and full-cell-development-frozen conditions are inductive: their inventories are fixed using development evidence. The two predefined factor policies encode expert priors over the full declared grammar domain. Full-cell–all is transductive because its KC inventory includes the exact held-out cells. It is useful as a resolution extreme, not as a fair inductive competitor.

The item-bank fingerprint and fixed-event fingerprint must match across all conditions. A condition changes only the selected policy, item–KC projection, Q-matrix, candidate learner histories, and predictions. This fixed-data requirement is checked before any pairwise result is interpreted.

5.2 Independent structural simulation oracle

The simulator maps each accepted item to a subset of ten data-generating features: finite form, finite agreement, perfect, progressive, and passive dependencies, negation, operator inversion, DO-support, central-modal structure, and imperative structure. Activation rules use the same observable cell and realisation evidence available before KC selection, but the features have a separate namespace and configuration. They are controlled simulator primitives, not candidate KCs and not proposed cognitive atoms.

For each of three profile settings, 60 learners are generated. Initial masteries are independent draws from $\mathrm{Beta}(2,6)$, $\mathrm{Beta}(2,2)$, or $\mathrm{Beta}(6,2)$; the corresponding learning rates are .06, .05, and .035. Item difficulty d_i is a deterministic hash of item identity mapped to $[-.8, .8]$. If $O(i)$ is the active oracle set and $m_{lo}^{(t)}$ is learner l's pre-event mastery, then

$$z_{li}^{(t)} = \frac{1}{|O(i)|} \sum_{o \in O(i)} \mathrm{logit}\Big(m_{lo}^{(t)}\Big) - d_i - .08\big(|O(i)| - 1\big), \tag{10}$$

$$p_{li}^{(t)} = .1 + .8\,\sigma\Big(z_{li}^{(t)}\Big). \tag{11}$$

After drawing $y_{li}^{(t)} \sim \mathrm{Bernoulli}(p_{li}^{(t)})$, each active oracle mastery is updated by

$$m' = \min\{.999, \max[.001, m + \eta_l(1-m)g_y]\}, \tag{12}$$

where $g_1 = .60$ and $g_0 = .35$. This is deliberately simple and monotonic; an incorrect response still creates a smaller learning opportunity.

The ordinary technical stream gives every learner two independently shuffled passes through the whole accepted bank and assigns chronological 60/20/20 train, validation, and test portions. Its public event record contains item, cell, timestamp, split, difficulty, and correctness, but no oracle or candidate KC fields. Initial states, probabilities, draws, and updates are held in separate private oracle artifacts for audit. Evaluated KC rules and counts are never simulator inputs.

Condition	Construction	Information category	Held-out behaviour
Selected structural	Greedy/backward cover of declared candidates	Inductive; development cells only	Frozen rules may activate on new combinations.
Factorised–development-frozen	Expert factor rules retained only with development item support	Inductive; expert hypothesis family	Supported reusable factors may transfer; unsupported factors are absent.
Full-cell–development-frozen	One KC for each exact development cell	Inductive exact-cell control	Held-out cells have zero active KCs and remain in evaluation.
Predefined factorised	Versioned expert rules for reusable marked factors	Expert prior over the full domain	Factor rules activate wherever their predicates match.
Factorised plus interactions	Predefined factors plus named structural interactions	Expert interaction prior	Both reusable factors and matching interactions can activate.
Full-cell–all	One KC for every exact cell in the inventory	Transductive inventory control	Every held-out cell is known as its own KC; no component reuse.

Table 3: Ontology conditions and their prior-information categories. The transductive control is descriptive and must not be compared with inductive methods as if it had the same information.

5.3 Development acquisition and frozen probes

The primary representation comparison uses a separate compositional protocol. Each learner receives two shuffled passes over development items only. The oracle state is then frozen. Every compositional-holdout and novel-feature item is probed once from that same post-development state; probe responses do not update it. Random draws are keyed by learner and item, making outcomes invariant to probe ordering.

For each ontology, candidate-KC opportunity counts, empirical histories, BKT mastery, and logistic features are learned from development acquisition and then frozen before all probes. Each held-out item is classified as *covered* (at least one KC activates), *fully development-supported* (all active KCs occurred on development items), *cold* (at least one active KC did not), or *uncovered*. A cold KC receives a fixed .5 prior. An uncovered event receives the same learner-global Beta-smoothed development success rate under every model and stays in the all-probe metric. Coverage and predictive performance are consequently reported separately; dismissing uncovered rows would reward narrow policies by changing the test set.

5.4 Technical KT baselines and evaluation

Three deliberately modest KT baselines test what the representation makes available. The empirical baseline predicts the mean Beta$(1, 1)$-smoothed pre-event success rate over active KCs. BKT uses fixed learning, guessing, and slipping parameters .04, .20, and .10; initial KC mastery is the clipped smoothed training rate. Logistic regression uses L_2 regularisation ($C = 1$, `lbfgs`, at most 300 iterations) and only pre-event information: learner-global success, active-KC success, mean log prior opportunities, fixed item difficulty, active-KC count, and one-hot KC indicators. No model reads the private oracle artifacts.

Log loss on fixed compositional probes is the primary metric because it is a proper scoring rule and penalises unjustified confidence. Brier score, AUC, accuracy at .5, and ten-bin expected calibration error are secondary. Every table must include item and event coverage, development-supported coverage, and cold-KC rate. Metrics are shown for all probes with the declared fallback, covered probes, and fully development-supported probes; the latter two are diagnostic subsets, not replacements for the primary test set.

Pairwise ontology contrasts use the same model technique and identical probe events. The estimand is mean event-level log loss for ontology A minus B. A paired learner bootstrap resamples learners with replacement 1,000 times, retaining all probes for each sampled learner, and reports the observed difference with a percentile 95% interval. The novel-feature split is reported separately and treated as exploratory because it contains one cell. The ordinary chronological benchmark remains a software sanity check, not the main answer to RQ3.

6 Current Evidence Status and Results Contract

No quantitative research result from RQ1–RQ3 is claimed in this revision. This is a substantive boundary rather than an unwritten section. The archived pilot outputs were generated under an earlier pipeline in which item pools and the response-generating representation could vary with the evaluated KC policy. The current method instead fixes both before policy selection. The old counts therefore answer a different, confounded question and cannot be carried forward selectively.

Table 4 distinguishes what has been verified from what still requires execution or review. At this revision, the complete automated suite passes 54 tests, including fixed-bank equality across ontologies, observable/oracle separation, development-only selection, zero-KC event retention, and non-updating probes. These tests establish implementation properties on controlled fixtures. They do not estimate normalisation quality or KT performance on the frozen EGP pilot.

The archived `base` directory does not pass the current run validator. It lacks the new `kc_selection` stage and the required normalisation, canonical-attrition, and repeated-item-diagnostic audit files. The expected external EGP file is also not distributed in the repository, and every manual checkpoint in the research-audit notebook remains unreviewed. Consequently:

- RQ1 has an executable measurement procedure but no reportable final mapping yield, transition counts, or reviewed error analysis;
- RQ2 has a declared candidate family and selector but no validated selected-policy artifact from the final pilot; and
- RQ3 has fixture-level proof of the fixed-data and frozen-state boundaries but no fresh pilot event stream or ontology comparison.

Table 5 specifies the outputs that must replace this status account after the first clean run. All fields are generated from versioned audit artifacts. Reporting them together prevents a high predictive score from obscuring low coverage, or a compact ontology from obscuring unsupported distinctions.

The acceptance gate for a quantitative revision is: a source digest match; a complete run of all six ontology conditions from the same accepted bank and event stream; `PASS` from the run validator; matching fixed-data fingerprints in every comparison; completion of the manual normalisation and item audits; and a result table populated directly from saved summaries. A failed gate remains visible and is not repaired by copying values from an obsolete run.

7 Interpretive Framework and Next Steps

Although results are pending, the implemented design clarifies what each eventual observation can and cannot mean.

Normalisation trades coverage for warranted claims. A low exact-cell yield would not by itself show poor annotation. The closed representation requires six scalar properties, whereas an EGP descriptor may intentionally state only one contrast. Retaining a partial mapping can be the correct outcome. RQ1 must therefore combine attrition rates with a manually reviewed taxonomy: unsupported completion, missed evidence, wrong scope, branching error, schema error, and genuine source underspecification. The right target is calibrated conservatism, not maximum downstream cell count.

Structural selection establishes sample support, not cognitive truth. If the selector retains a candidate, the result means that its activation pattern contributes to the declared development obligations after equivalence and scope checks. It does not show that learners acquire that candidate as an indivisible skill. Conversely, a rejected candidate may be meaningful outside the small inventory but unsupported or indistinguishable within it. The selection trace, equivalence classes, and holdout activation patterns are at least as important as the final KC count.

Coverage is part of predictive performance. Exact-cell KCs offer high structural specificity but cannot transfer learner evidence to a new cell unless that cell was admitted in advance. Reusable factors can cover new combinations, but may average evidence across cases whose difficulty or learning dynamics differ. Interactions can express residual dependencies while increasing sparsity. The fixed probe protocol exposes this trade-off by showing all-probe loss beside coverage, cold-start, and development-supported subsets. No policy can improve its headline metric by discarding the items it fails to annotate.

The simulation is a mechanism check. The independent oracle removes the circularity of generating responses from the candidate KCs under evaluation. It still favours representations capable of recovering its particular factorised structural regularities. A policy that transfers well in this world has demonstrated information reuse under a

Evidence layer	Current status	Consequence for this report
Executable specifications and boundary tests	Available; 54/54 tests pass	The declared transformations, leakage guards, frozen-probe behaviour, and six-policy fixed-data fixture can be described as implemented.
Archived `runs/base` output	Predates the final pipeline; current validation fails four gates	Its normalisation, item, policy, and KT counts are not reported as results of the present method.
External EGP snapshot	Consult-only and intentionally not redistributed	A fresh run requires the separately held file matching the declared SHA-256 and 1,222-record count.
Model-assisted manual audit	Audit notebook checkpoints remain NOT REVIEWED	Automated validity and repeat diagnostics cannot yet be promoted to expert linguistic validation.
Current full experimental run	Not yet generated and validated	RQ1–RQ3 quantitative estimates and ontology rankings remain pending.

Table 4: Evidence boundary at the report revision. "Implemented" and "empirically supported" are deliberately treated as different statuses.

Question	Required reported evidence	Interpretation guard
RQ1: normalisation	Phase-1 and final status counts; Phase-2 transition table; admitted source edges and unique cells; repeat agreement by decision type; manually reviewed error categories and examples	Exact-cell yield measures conservatism and coverage, not correctness; automated repeats are not inter-annotator agreement.
RQ2: structural support	Candidate support and scope diagnostics; equivalence classes; unmet and satisfied obligations; complete selection trace; retained KCs; development versus holdout activation; Q-matrix coverage, density, width, support, and redundancy for every condition	Deterministic representative choice does not distinguish equivalent hypotheses; a feasible inclusion-minimal cover is not a global or cognitive optimum.
RQ3: compositional transfer	Identical item/event fingerprints; covered, fully supported, cold, and uncovered rates; all-probe log loss; secondary metrics; paired learner-bootstrap differences and 95% intervals; novel-feature results separately	Comparisons are conditional on the declared synthetic oracle and baseline models; subset metrics cannot replace the all-probe metric.

Table 5: Result-reporting contract for the first fresh validated run. These fields replace ad hoc selection of favourable counts or metrics.

known mechanism, not correspondence to human grammar acquisition. Robustness to seeds and alternative declared oracles would test engineering dependence; human learner sequences are required for educational validity.

Path to tutoring dialogue. TSCCv2 motivates the project but is not an evaluation dataset in the present study. Moving from controlled items to dialogue requires three new layers: identifying which turns are genuine assessment opportunities, assigning context-sensitive correctness, and allowing an utterance to instantiate multiple grammatical phenomena. These labels should be created and audited without using downstream KT scores to decide the ontology. The controlled benchmark can then serve as a unit-tested bridge, while real dialogue supplies the independent learner evidence the current simulation lacks.

The immediate completion sequence is therefore empirical rather than editorial: restore the verified consult-only source, run and validate the reference condition, freeze its pre-selection outputs, execute the five remaining policy variants through KT, complete the manual audits, and populate Table 5. Only then should the discussion be revised to answer the three RQs with estimates rather than expectations.

8 Reproducibility and Audit Trail

Scientific choices are stored as data rather than hidden in the report text. The canonical schema, normalisation prompts, transition rules, lexical frames, item-bank design, diagnostic acceptance rule, fold, candidate family, selection obligations, predefined policies, oracle, and KT parameters are independently versioned. Executors validate these artifacts before applying them. Appendix A maps each conceptual component to its repository location.

Every run records the resolved experiment, random seed, external source identity, Git commit and dirty state, and signatures of upstream inputs. Stage reuse is opt-in: a child experiment names a parent and starting stage, and the runner refuses to copy earlier outputs if any recursively referenced input has changed. There is no implicit cache. Model-assisted stages additionally retain prompts, raw and parsed responses, invocation metadata, and errors for each attempt.

The source contract declares a SHA-256 beginning `e38c4f959f188150...` and ending `c2488486cd`, and a total of 1,222 records; the full digest remains in the executable experiment file. Because the snapshot is consult-only, it is located through `GRAMMAR_KT_EGP_SOURCE` or a declared data-root variable and is not redistributed. A digest or record-count mismatch is fatal.

The working model configuration currently requests `gpt-5.6-sol` at medium reasoning effort in a fresh read-only context for both normalisation and item diagnostics. The service snapshot and all decoding parameters are not pinned. Exact external re-execution is therefore not guaranteed even with identical local files. Retaining every response and repeating selected units makes this limitation inspectable; it does not remove it.

The final verification boundary comprises four command-level checks:

- execute the complete `unittest` discovery suite with `PYTHONPATH=src`;
- generate `base` with `scripts/run.py` and require a clean result from `scripts/validate.py`;
- run each child condition explicitly from `kc_selection`; and
- use `scripts/compare.py` at the compositional stage to verify fixed inputs and calculate paired contrasts.

Equivalent runs are required for the other conditions in Table 3. Inspection commands trace a descriptor, cell, item, Q-matrix row, ordinary event, or probe through its saved evidence. Automated tests verify implementation and leakage boundaries; the research audit notebook separately records linguistic and methodological review. A green test suite does not override an unreviewed notebook or a failed run validator.

9 Limitations and Ethics

Representation and sampling. The ontology is a deliberately narrow engineering representation of single-clause verbal morphosyntax. It omits many constructions, meanings, and discourse dependencies relevant to grammar learning. Its values need not be cognitively atomic. The 139 descriptors form a purposive pilot with explicit negative controls, not a representative sample of the EGP, English grammar, or learner needs. Only complete mappings contribute canonical cells, so every downstream result is conditional on normalisation attrition.

Automated judgement. One unpinned model configuration performs normalisation and item diagnostics. Eight mapping repeats and five diagnostic repeats are small consistency probes, not estimates of inter-annotator reliability. Structural validation can detect forbidden values and transformations but cannot establish that a mapping captures the source meaning or that an item is pedagogically natural, unbiased, and uniquely answerable. Qualified human review and adjudication remain necessary.

Controlled language and item validity. Ten predicate frames and templated transformations make derivations auditable but sharply restrict lexical, pragmatic, and register variation. The benchmark is closer to a structural test harness than an authentic assessment. Success may reflect regularities of the templates, and a rule that works there may fail in free production or dialogue.

Synthetic learners. The oracle assumes independent initial dimensions, monotonic learning, fixed profile-level rates, deterministic item difficulty, and a simple response equation. It has no forgetting, transfer beyond shared oracle features, misconceptions, response strategies, affect, or dialogue context. Profile labels refer only to Beta-distribution settings and must not be read as demographic, diagnostic, or stable ability groups. Predictive comparisons measure recovery and reuse within this controlled world, not human learning.

Evaluation scale. There are only seven compositional holdout cells and one novel-feature cell. Although each synthetic learner supplies repeated fixed probes, the diversity of grammatical generalisation targets remains small. Learner-level bootstrap intervals quantify sampling variability in the simulated population, not uncertainty about ontology design, source annotation, or generalisation to other grammar domains. Multiple oracle specifications and human data would be needed for those claims.

Ethical and licensing considerations. No human-participant or learner records are used in this pilot. This reduces privacy risk but also removes the population whose learning the eventual system is meant to model. The EGP snapshot must be handled under its licence; the artifact records identifiers and verification material rather than redistributing the source. Generated items and inferred KCs should not be used for teaching, placement, or learner profiling without expert review, accessibility and bias assessment, and validation with consented learner data.

10 Conclusion

Grammar–KT makes the normally implicit path from curriculum prose to KT content annotations explicit and testable. The current framework conservatively maps EGP descriptors into six-dimensional grammar cells, generates a split-independent and KC-independent item bank, fixes outcomes through a separate structural oracle, selects structural hypotheses using development evidence only, and compares frozen representations on identical non-updating probes. This design removes the principal circularity of the earlier pilot: an evaluated ontology can no longer choose its own items or generate its own answers.

The writeable methodological contribution is now complete, but the empirical claim is intentionally not. Archived counts from the earlier design are not valid results for the corrected pipeline. A verified source, a fresh six-condition run, passing audits, and expert review are the remaining gates. When those are met, the predeclared result contract will support bounded answers about normalisation reliability, structural support, Q-matrix coverage, and synthetic compositional transfer—not claims that the resulting KCs are established units of human grammar learning.

References

Ghodai Abdelrahman, Qing Wang, and Bernardo Pereira Nunes. 2023. Knowledge tracing: A survey. *ACM Computing Surveys*, 55(11):1–37.

Semere Kiros Bitew, Johannes Deleu, A. Seza Doğruöz, Chris Develder, and Thomas Demeester. 2023. Learning from partially annotated data: Example-aware creation of gap-filling exercises for language learning. In *Proceedings of the 18th Workshop on Innovative Use of NLP for Building Educational Applications*, pages 598–609. Association for Computational Linguistics.

Andrew Caines, Helen Yannakoudakis, Helen Allen, Pascual Pérez-Paredes, Bill Byrne, and Paula Buttery. 2022. The teacher-student chatroom corpus version 2: More lessons, new annotation, automatic detection of sequence shifts. In *Proceedings of the 11th Workshop on NLP for Computer Assisted Language Learning*, pages 23–35. LiU Electronic Press.

Hao Cen, Kenneth R. Koedinger, and Brian Junker. 2006. Learning factors analysis: A general method for cognitive model evaluation and improvement. In *Intelligent Tutoring Systems*, volume 4053 of *Lecture Notes in Computer Science*, pages 164–175. Springer.

Sophia Chan, Swapna Somasundaran, Debanjan Ghosh, and Mengxuan Zhao. 2022. AGReE: A system for generating automated grammar reading exercises. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 169–177. Association for Computational Linguistics.

Albert T. Corbett and John R. Anderson. 1995. Knowledge tracing: Modeling the acquisition of procedural knowledge. *User Modeling and User-Adapted Interaction*, 4(4):253–278.

Council of Europe. 2020. *Common European Framework of Reference for Languages: Learning, Teaching, Assessment—Companion Volume*. Council of Europe Publishing, Strasbourg.

Peng Cui and Mrinmaya Sachan. 2023. Adaptive and personalized exercise generation for online language learning. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10184–10198. Association for Computational Linguistics.

Jimmy de la Torre and Chia-Yi Chiu. 2016. A general method of empirical Q-matrix validation. *Psychometrika*, 81(2):253–273.

Shuyan Huang, Alexander Scarlatos, Jaewook Lee, and Andrew Lan. 2026. Interpretable difficulty-aware knowledge tracing in tutor-student dialogues. In *Proceedings of the 21st Workshop on Innovative Use of NLP for Building Educational Applications*, pages 612–623. Association for Computational Linguistics.

Kenneth R. Koedinger, Albert T. Corbett, and Charles Perfetti. 2012. The knowledge-learning-instruction framework: Bridging the science-practice chasm to enhance robust student learning. *Cognitive Science*, 36(5):757–798.

Anne O'Keeffe and Geraldine Mark. 2017. The English Grammar Profile of learner competence: Methodology and key findings. *International Journal of Corpus Linguistics*, 22(4):457–489.

Panagiotis Pagonis, Kai Hartung, Di Wu, Munir Georges, and Sören Gröttrup. 2024. Analysis of knowledge tracing performance on synthesised student data. *arXiv preprint arXiv:2401.16832*.

Zachary A. Pardos and Anant Dadu. 2018. dAFM: Fusing psychometric and connectionist modeling for Q-matrix refinement. *Journal of Educational Data Mining*, 10(2):1–27.

Philip I. Pavlik, Hao Cen, and Kenneth R. Koedinger. 2009. Performance factors analysis—a new alternative to knowledge tracing. In *Proceedings of the 14th International Conference on Artificial Intelligence in Education*, volume 200 of *Frontiers in Artificial Intelligence and Applications*, pages 531–538.

Chris Piech, Jonathan Bassen, Jonathan Huang, Surya Ganguli, Mehran Sahami, Leonidas Guibas, and Jascha Sohl-Dickstein. 2015. Deep knowledge tracing. In *Advances in Neural Information Processing Systems 28*, pages 505–513.

Alexander Scarlatos, Ryan S. Baker, and Andrew Lan. 2025. Exploring knowledge tracing in tutor-student dialogues using LLMs. In *Proceedings of the 15th International Learning Analytics and Knowledge Conference*, pages 249–259. ACM.

Alexander Scarlatos, Jaewook Lee, Simon Woodhead, and Andrew Lan. 2026. Simulated students in tutoring dialogues: Substance or illusion? In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 42349–42385. Association for Computational Linguistics.

Megha Srivastava and Noah Goodman. 2021. Question generation for adaptive education. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 2: Short Papers)*, pages 692–701. Association for Computational Linguistics.

Daniela Verratti-Souto, Nelly Sagirov, and Xiaobin Chen. 2025. NLP-powered quantitative verification of the English Grammar Profile's structure-level assignment. *Annual Review of Applied Linguistics*, 45:109–130.

A Artifact Map

Component	Declaration, executor, and retained evidence
Source	`experiments/base.yaml`; source sampling manifests and annotation units; `source.py`; selection and digest audit
Normalisation	Phase prompts, rules, schema, and backend; `normalisation.py`; per-unit calls and `reliability.json`
Canonical representation	`grammar_schema.yaml`; `canonical.py`; cells, source edges, and attrition audit
Realisation	Ten-frame lexicon and rules; `realisation.py`; specifications and derivations
Item bank	Bank design, transformation, and diagnostic acceptance; `items.py`; intrinsic bank, fingerprint, and repeat reliability
Canonical fold	`reference_v0.json`; runtime join and exact-inventory audit
Structural selection	Candidates, obligations, and selector; `kc_candidates.py`; diagnostics, equivalence classes, trace, and frozen policy
KC policies	Factorised, interaction, full-cell, and generated policies; `kc.py`; item projection and inventory
Q-matrix	Frozen projection; `qmatrix.py`; edge list, matrix, and diagnostics
Simulation	Structural oracle, profiles, protocol, and seed; `simulation.py`; fixed events, private evidence, fingerprints, and probes
KT evaluation	Baseline and bootstrap settings; `kt.py`; histories, predictions, coverage, metrics, and comparisons

Table 6: Conceptual choices and the repository artifacts that make them inspectable. Paths are relative to the project root.

B Fresh-Run Completion Checklist

The following checklist is the remaining route from this methodological draft to a quantitative report.

1. Restore the consult-only EGP JSONL outside version control and verify its declared digest and record count before allowing model calls.

2. Run base from the source stage and require a clean `scripts/validate.py base` result. Do not use archived outputs to fill missing audits.

3. Complete every checkpoint in `notebooks/research_audit.ipynb`, including Phase-1/Phase-2 transitions,

source–cell traces, item naturalness and ambiguity, fold logic, oracle separation, and probe freezing. Record reviewer, decision, and adjudication evidence.

4. From the same validated pre-selection parent, execute `kc_selected_structural`, `kc_factorized_dev_frozen`, `kc_full_cell_dev_frozen`, `base`, `kc_interactions`, and `kc_full_cell`. Validate every child run.

5. Compare intrinsic item-bank, ordinary event-stream, acquisition-stream, and probe-stream fingerprints across all six conditions. Stop if any differ.

6. Populate RQ1 from `normalisation/reliability.json` and `canonical/audit.json`; RQ2 from `kc_selection/evaluation.json` and Q-matrix diagnostics; and RQ3 from `kt/compositional/metrics.json` plus paired comparison artifacts. Preserve zero-KC rows and report denominators.

7. Re-run the complete test suite and compile the report from a recorded commit. Archive the resolved experiment files and the exact tables used in the PDF.

The checklist deliberately ends with table population rather than narrative selection. Any unexpected failure, mapping disagreement, rejected item, cold KC, or uncovered probe is part of the result and should remain visible.